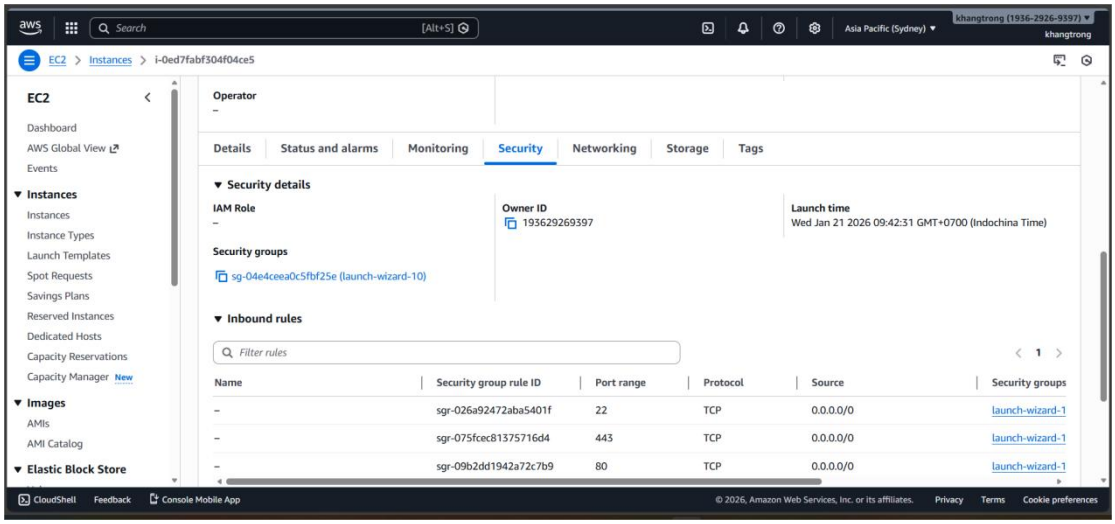



Image 1. EC2 security group

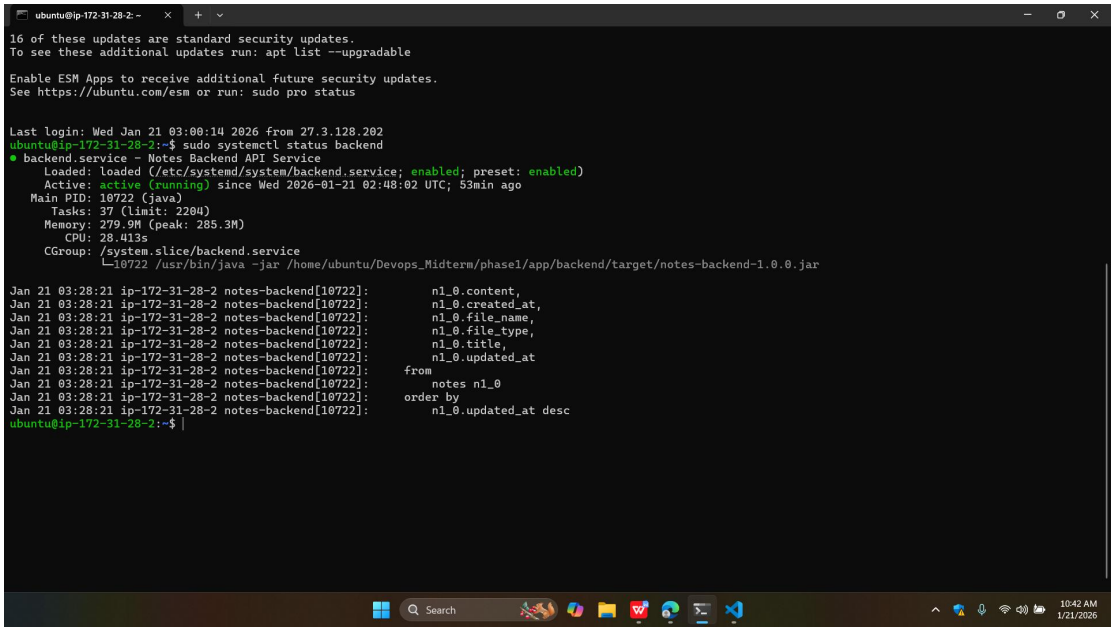


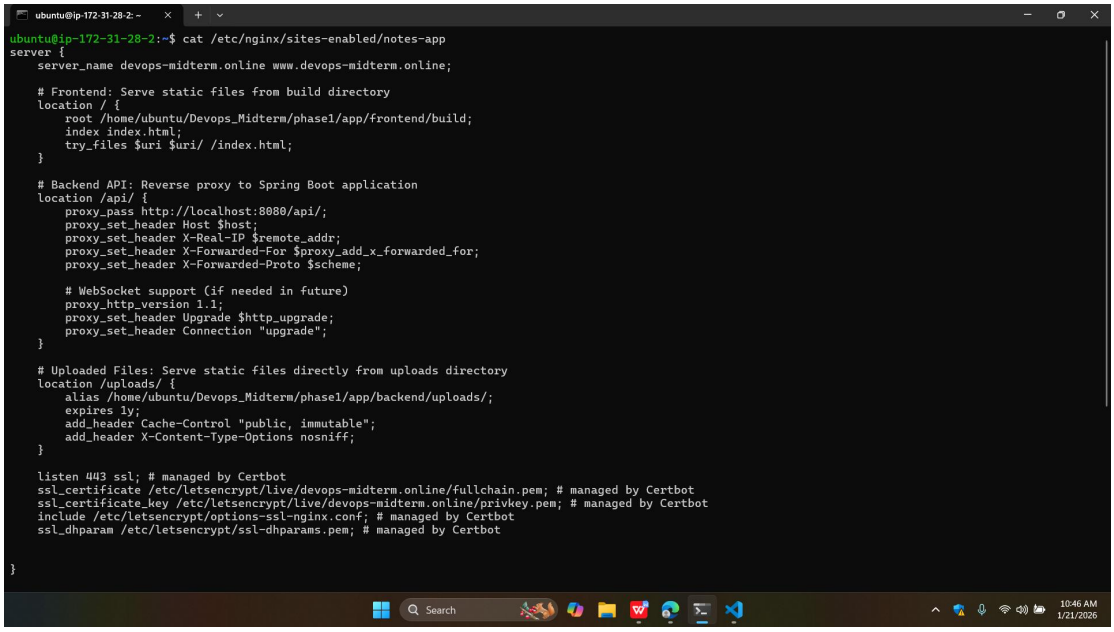

Image 2. Backend service status running on Ubuntu server using systemctl

```
ubuntu@ip-172-31-28-2:~$ sudo systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
     Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
     Active: active (exited) since Wed 2026-01-21 02:45:31 UTC; 58min ago
   Main PID: 6373 (code=exited, status=0/SUCCESS)
        CPU: 2ms

Jan 21 02:45:31 ip-172-31-28-2 systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
Jan 21 02:45:31 ip-172-31-28-2 systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
ubuntu@ip-172-31-28-2:~$
```

Image 3. PostgreSQL service status managed by systemd on Ubuntu 24.04

```
ubuntu@ip-172-31-28-2:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-01-21 03:02:59 UTC; 41min ago
       Docs: man:nginx(8)
    Process: 10983 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 10986 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 10988 (nginx)
      Tasks: 3 (limit: 2204)
     Memory: 4.0M (peak: 5.7M)
        CPU: 137ms
     CGroup: /system.slice/nginx.service
             ├─10988 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─11010 "nginx: worker process"
             └─11011 "nginx: worker process"

Jan 21 03:02:59 ip-172-31-28-2 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jan 21 03:02:59 ip-172-31-28-2 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-28-2:~$
```

Image 4. Nginx service status managed by systemd on Ubuntu 24.04



```
ubuntu@ip-172-31-28-2:~$ cat /etc/nginx/sites-enabled/notes-app
server {
    server_name devops-midterm.online www.devops-midterm.online;

    # Frontend: Serve static files from build directory
    location / {
        root /home/ubuntu/Devops_Midterm/phase1/app/frontend/build;
        index index.html;
        try_files $uri $uri/ /index.html;
    }

    # Backend API: Reverse proxy to Spring Boot application
    location /api/ {
        proxy_pass http://localhost:8080/api/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;

        # WebSocket support (if needed in future)
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
    }

    # Uploaded Files: Serve static files directly from uploads directory
    location /uploads/ {
        alias /home/ubuntu/Devops_Midterm/phase1/app/backend/uploads/;
        expires 1y;
        add_header Cache-Control "public, immutable";
        add_header X-Content-Type-Options nosniff;
    }

    listen 443 ssl; # managed by Certbot
    ssl_certificate /etc/letsencrypt/live/devops-midterm.online/fullchain.pem; # managed by Certbot
    ssl_certificate_key /etc/letsencrypt/live/devops-midterm.online/privkey.pem; # managed by Certbot
    include /etc/letsencrypt/options-ssl-nginx.conf; # managed by Certbot
    ssl_dhparam /etc/letsencrypt/ssl-dhparams.pem; # managed by Certbot


}
```

Image 5. Nginx configuration file defining reverse proxy routing, static file serving, upload directory mapping, and HTTPS configuration HTTPS certificate status (browser lock icon, certbot output, etc.).

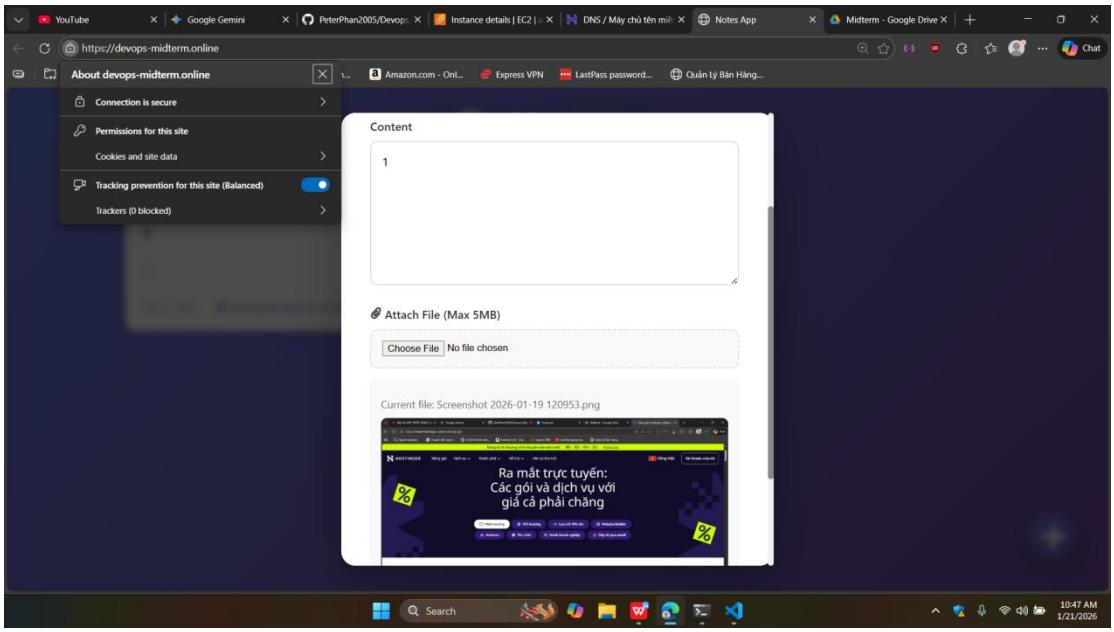



Image 6. Website secured with HTTPS

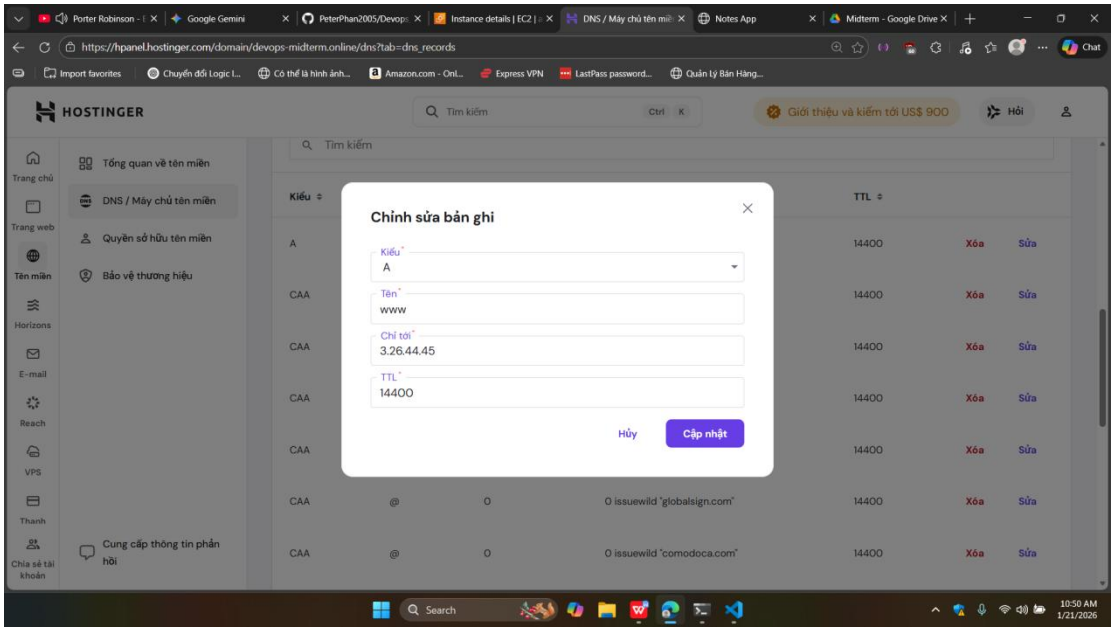



Image 7. Configuration of the www domain record

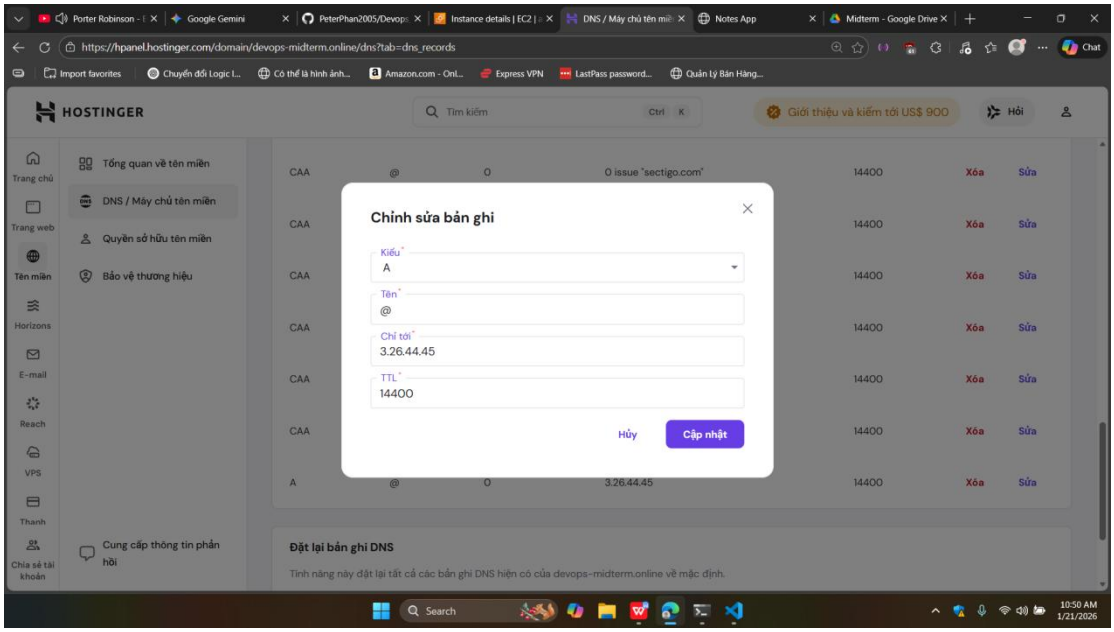



Image 8. DNS A record configuration for the root domain (@)